

# Signals

**Willi, Mojca & Hans**

context **2025** meeting

# From beamer to lamps

- We wanted Tove's BachoT<sub>E</sub>X 'flags' presentation to have some weird properties.
- I was working on a large 65 inch screen on 3 meter distance due to an eye issue.
- I had some hue color lamps laying around unused and the home/office lighting is driven by a few hue hubs already.
- Control of lamps is actually done by a Lua script (server) run by LuaT<sub>E</sub>X, a way to overcome limitations (and performance issues).
- So, controlling lamps was relatively easy. Making them respond to a change in pdf page not that hard.
- That in return triggered using lamps to indicate a ConT<sub>E</sub>Xt run state and result.

# From lamps to leds

- At BachoT<sub>E</sub>X we played a bit with a led strip (on a hue led controller) that Mojca gave me.
- That demonstrated that instead of (large) lamps we could go smaller.
- We used 30 leds and therefore could play with configurations. Eventually 24 leds were the optimum (1 2 3 4 6 8 12).
- By putting them in a circle one gets a kind of clock (watch).
- We use colors: blue (busy), yellow (current), red (error), orange (problem), green (okay).
- We can work on segments (24/8), quadrants (24/4), or squids (24/1). The first two report per run (if more are needed), the later the current run with page indicator.

# From leds to gadget

- Mojca suggested to make a dedicated device, and gave me a Raspberry Pico to play with.
- The quadrant, segment, squid model was ‘perfected’ in an experimental setup.
- We still support hue, we can also respond to http requests but the focus is now on fast serial communication.
- Controlling the gadget (aka ConT<sub>E</sub>Xt watch, as it is round) happens from the context script and in squid mode also from the shipout routine.
- Serial communication is built into LuaMetaT<sub>E</sub>X, and it works okay on Windows, linux and os-x.

# From gadget to tool

- We can see progress and result state using quadrant, segment, or squid mode. The documents top line indicates what has to be used.
- In squid mode we can also get feedback about e.g. overfull boxes, missing characters, options that slow down a run, or potentially interfering third party code.
- In segment mode we can show the state of 8 parallel jobs, this is used when running the test suite.
- You can control the device by sending commands and thereby integrate it in workflows. We use this when generating the distribution.
- Future versions can have a clock mode. We already can count (numbers) and do some simple text.

# How it fits in

- It all fits in a feedback/runner model. When a run goes bad, one gets an error document.
- But instead one can get a QR code as result, encoding the error (there is a limited set so basically they are pictograms).
- Using leds (color and/or position) in practice is a faster feedback than watching and reading a log file.
- It also marks the 20<sup>th</sup> anniversary of LuaT<sub>E</sub>X, we've come a long way and still finds ways to extend it.
- The software is there but can be upgraded as we go. There's also a manual and a helper script in the distribution.
- As the gadget can be controlled by a script you can also use it otherwise, but as usual, we welcome ideas by users.

# Dejavu

- I played a lot with a RCA 1802 dev kit when was a kid myself. It was one of nicer cpu designs, quit symmetrical with static memory and single step capabilities. Think of the kind of chips that currently travel the edges of our solar system.
- Writing games and such in 512 bytes mem (or was it 256) is a challenge.
- Inventing wheels is part of it. One learns a lot: interrupts, self adapting functions, figuring out how to save to and read from tape, dma and sensors.
- When we started with Lua<sub>T</sub><sub>E</sub>X, at some point Mojca send me a esp micro controller running Lua. It brought back memories.
- Years later Harald made a laser cut Con<sub>T</sub><sub>E</sub>Xt logo controlled by a rpi and later Willi made an enclosure with a next generation embedded esp micro controller.

- Today we show a gadget that has yet another generation of that esp (although we actually started with a rpi pico).
- For a moment think back to when T<sub>E</sub>X was written: this tiny chip has more cpu cycles, flash, plenty static and optionally extra memory and quite some IO features, it can run original T<sub>E</sub>X faster than the machines on which he program was written.

# The workshop



- If there is interest we will make more boards that can be mounted in a clock encasing with a 3D printed enclosure.
- But ... at this meeting you will get the collectors item: a high quality wooden Willi designed stand with the first official hard- and software release.
- But you need to work a bit on it, welcome to:  
*The traditional ConT<sub>E</sub>Xt workshop event!*